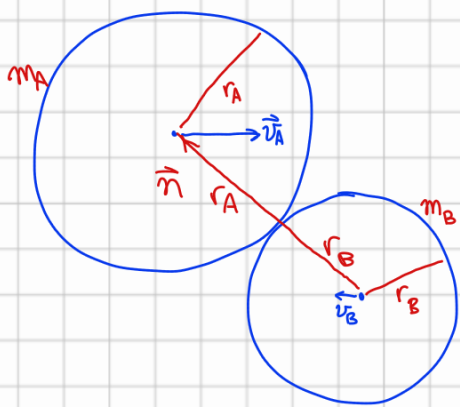


Ball to ball collision.



$$\vec{n} = \vec{x}_A - \vec{x}_B$$

Key observation: momentum can only be transferred in the $\vec{n}$ direction.

$\Rightarrow$ velocities in orthogonal directions are preserved.

Conservation of momentum in $\vec{n}$ direction means that if $v_{A,\hat{n}}$ changes by δv , then $v_{B,\hat{n}}$ changes by $-\frac{m_A}{m_B} \delta v$.

However, there is still a degree of freedom arising from the degree of elasticity in the collision.

• Fully plastic collision (the balls stick):

$$v_{A,\hat{n}}^{\text{new}} = v_{B,\hat{n}}^{\text{new}} \Rightarrow v_{A,\hat{n}} + \delta v = v_{B,\hat{n}} - \frac{m_A}{m_B} \delta v$$

$$\Rightarrow \left(1 + \frac{m_A}{m_B}\right) \delta v = v_{B,\hat{n}} - v_{A,\hat{n}}$$

$$\Rightarrow \delta v = \frac{m_B}{m_A + m_B} (v_{B,\hat{n}} - v_{A,\hat{n}})$$

$$= \frac{1/m_A}{1/m_A + 1/m_B} (v_{B,\hat{n}} - v_{A,\hat{n}})$$

• Fully elastic collision: Energy is conserved:

$$\cancel{m_A v_{A,\hat{n}}^2} + \cancel{m_B v_{B,\hat{n}}^2} = m_A (v_{A,\hat{n}} + \delta v)^2 + m_B (v_{B,\hat{n}} - \frac{m_A}{m_B} \delta v)^2$$

$$= \cancel{m_A v_{A,\hat{n}}^2} + 2m_A v_{A,\hat{n}} \delta v + m_A \delta v^2$$

$$+ \cancel{m_B v_{B,\hat{n}}^2} - 2m_A v_{B,\hat{n}} \delta v + m_B \left(\frac{m_A}{m_B}\right)^2 \delta v^2$$

$$0 = \cancel{2m_A \delta v (v_{A,\hat{n}} - v_{B,\hat{n}})} + \cancel{m_A \delta v^2} \left(1 + \frac{m_A}{m_B}\right)$$

$$\delta v \left(\frac{m_A + m_B}{m_B}\right) = 2(v_{B,\hat{n}} - v_{A,\hat{n}})$$

$$\delta v = 2 \frac{m_B}{m_A + m_B} (v_{B,\hat{n}} - v_{A,\hat{n}})$$

which is exactly 2x the velocity transfer in the plastic collision.

• Generalization: Define elasticity $0 \leq e \leq 1$,

then

$$\delta v = (1+e) \frac{m_B}{m_A + m_B} (v_{B,\hat{n}} - v_{A,\hat{n}})$$